



Title: Human vs. AI Forecast Challenge: A Comparative Forecast of USD/BDT Exchange Rate

Course: International Financial Management (FIN4810)

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Batch: BBA 10

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Date: 09/10/2025

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Section 1: Human Forecast (USD/BDT – 4-Month Outlook)

Forecasting the future exchange rate between the U.S. dollar and Bangladeshi taka entails a collusion of economic theory with current market conditions. For this report, I've collected all the relevant information from credible primary sources such as Bangladesh Bank, IMF & Trading Economics and I have used 3 basic frameworks of International Financial Management which include Purchasing Power Parity (PPP), Interest Rate Parity (IRP) & the Balance of Payments. for the analysis preparation.

Theoretical Foundation

If a country's inflation rate is higher than that of its trading partners, the PPP theory says the currency should lose value. Since inflation in Bangladesh is already running at around 9–10%, and U.S. inflation seems locked near the range of 3–4% (Trading Economics, 2025, n.d.), this discrepancy explains why the taka is placed under depreciation pressure. For example, the policy interest rate in Bangladesh is 8.5% and the U.S. Federal funds rate is 5.5%. By using the IRP equation, the expected forward rate is 122.9 BDT/USD with an estimated annual depreciation of about 2.8%. However, given that Bangladesh has partial capital controls and convertibility constraints, the full IRP does not directly apply to the real market . Finally, BOP theory connects changes in exchange rates to a country's external balance. The reserves for Bangladesh are at approximately USD 19.5 billion alongside a current account deficit of nearly USD 3.8 billion (Bangladesh Bank, 2025). Import growth continues to outpace exports and put additional pressure on the foreign exchange market. While remittance flows provide some support, they have not been sufficiently large to offset the increasing deficit.

Macroeconomic Context and Scenarios

Currently, Bangladesh Bank manages the currency utilizing the crawling peg or managed float approach, complemented with stabilization measures from the IMF. Even so, the pressures which can be admitted are short-term in nature, such as LC restrictions on imports, higher world fuel prices, and reduced competitiveness of exports. Looking historically, around 1–1.5% monthly volatility is realistic.

There are three possible scenarios:

- Positive: Ongoing IMF support with exports picking up operations and steady remittance flows keeps the exchange rate hovering around 120 BDT/USD.
- Realistic: Ongoing depreciation continues as inflation remains stubborn and import prices are high.
- Negative: Reserve amounts go below USD 18 billion with external shocks taking place, the rate could overshoot to above 124 BDT/USD.

Information and Inference

In early October 2025, Bangladesh Bank has set a reference rate of 119.5 BDT per USD (Bangladesh Bank, 2025). Keeping inflation, importing capacity and reserve levels in mind, I expect the taka to depreciate about 1% a month with a minor uptick in January as remittance flows increase.

Human Forecast Table

Month	Projected USD/BDT	Basis/Context
Oct 2025	119.5	BB reference, current spot rate
Nov 2025	120.7	Inflation gap, rising import and LC pressure
Dec 2025	122.0	Peak import season, ongoing reserve strain
Jan 2026	121.5	Seasonal remittance inflows, IMF-driven stability

In summary, I take this as a forecast of continued gradual depreciation for the next three months and a brief stabilization in January, as remittance flows improve and the IMF aligns some of their reforms on the ground.

Section 2: AI Forecast (ChatGPT Free Version)

Prompt Used (Condensed)

“Using data from Bangladesh Bank, Trading Economics, and IMF projections, provide a four-month forecast for USD/BDT. Apply PPP, IRP, and BOP logic in your reasoning.”

Summary of AI Forecast

The AI-generated forecast showed a slower and smoother trend of depreciation. It thought that Bangladesh's monetary policy and IMF reforms would keep the exchange rate from changing too much. But it didn't take into account things that were happening on the ground, like late LC settlements, reserve stress, or seasonal import surges (Trading Economics, 2025).

AI Forecast Table

Month	Projected USD/BDT	AI Rationale
Oct 2025	119.8	Slight depreciation from inflation differential
Nov 2025	120.5	Moderate import pressure, but stable policy outlook
Dec 2025	121.2	Seasonal weakening, controlled through policy measures
Jan 2026	120.9	Improved remittance flows, IMF-driven confidence

Overall, the AI forecast reflects global stability assumptions but does not capture the short-term local volatility typical of Bangladesh's managed exchange system (IMF 2025, n.d.)

Section 3: Comparison and Discussion

1. Which forecast was more difficult to produce and why?

To make the human forecast, data had to be gathered from many places and three different theoretical frameworks had to be used together. It also required making decisions about how the local economy worked, which is not something that can be easily automated. The AI prediction, on the other hand, was made right away, but it didn't take into account things that are specific to each country, like limits on imports and running out of reserves.

2. How did the forecasts differ, and which is more credible?

According to the human forecast, the currency's value would fall more quickly in the near-term, mainly in December, due to pressures from imports and reserves, followed by a correction in January. The AI forecast showed changes that took place more slowly with less degree of variation. In Bangladesh, the human forecast is considered to be more accurate because it takes real events from the Balance of Payments into account, along with local monetary

policies. The AI forecast does function well, but it would typically miss these limits based on broader macroeconomic assumptions.

3. Can SMEs in Bangladesh benefit from such tools and processes?

Certainly, small and medium-sized businesses can derive real value from implementing these strategies. Consider, for example, a business on Facebook selling imported beauty or electronic products. That business can use forecasts of exchange rates to inform decisions on when to pay the supplier or price the goods. Artificial intelligence tools provide fast and useful directional guidance, while human thinking contextualizes observed market fluctuations. Discounting one or both of these methods would limit the ability of SMEs to improve their financial risks management decision-making.

Appendix: IRP Calculation and BOP Justification

Interest Rate Parity Analysis (IRP)

- According to Bangladesh Bank (2025), the country's policy interest rate is 8.50%.
- According to the Federal Reserve Bank of St. Louis (2025), the Federal Funds interest rate in the United States is 5.50%.
- The exchange rate = 119.5 BDT to the USD

Formula:

$$F = S \times \{(1+i_{\text{USD}})/(1+i_{\text{BDT}})\}$$

$$F = 119.5 \times (1.055/1.085) = 119.5 \times 1.0284 = 122.9$$

Explanation : This highlights an average depreciation of 2.8% a year or roughly 0.23% a month. While it does support the depreciation forecast, the IRP is not entirely applicable to Bangladesh because of restrictions on markets and non-convertibility.

Balance of Payments (BOP) Analysis

Bangladesh has approximately USD 19.5 billion in foreign exchange reserve, and the country is facing a current account deficit of USD 3.8 billion, which is putting a strain on the supply of dollars. Remittance inflows of nearly USD 21 billion per year provide some support, particularly in the January timeframe, which reportedly would support the modest recovery anticipated for January.

ChatGPT Resource Link - <https://chatgpt.com/share/68e6ed1a-cb4c-800d-8f08-d3081b6d522d>

References

- Bangladesh Bank 2025. (n.d.). *Bangladesh Bank Exchange Rate of Taka*. Retrieved from <https://www.bb.org.bd/en/index.php/econdata/exchangerate>
- FEDERAL RESERVE BANK of ST LOUIS. (n.d.). Retrieved from FRED: <https://fred.stlouisfed.org>
- IMF 2025. (n.d.). *IMF Datamapper Bangladesh*. Retrieved from <https://www.imf.org/external/datamapper/profile/BGD>
- Trading Economics, 2025. (n.d.). *Bangladesh Inflation Rate MoM*. Retrieved from <https://tradingeconomics.com/bangladesh/inflation-rate-mom>